



Planning (PLAN)

PLAN Overview

Required PA Information

Intent

Develops plans to describe what is needed to accomplish the work within the standards and constraints of the organization.

Value

Optimizes cost, functionality, and quality to increase the likelihood of meeting objectives.

Additional Required PA Information

Planning should include an approach, activities, and actions needed for the following:

- Identifying and addressing requirements including quality, functionality, customers, users, etc.
- Developing budgets and schedules based on estimates
- Identifying resource demands and obtaining the necessary resource capacity and availability
- Identifying the appropriate set of stakeholders and tasks
- Managing risks and opportunities
- Developing and keeping the project plan updated to reflect how to perform the work

Plans also describe:

- The work to be performed
- Applicable organizational set of standard processes, assets, and tailoring guidelines
- Dependencies
- Who performs the work
- Relationships with other plans
- Stakeholders and their role

Capacity and availability management activities can be performed at different levels of the organization and applied to any type of work. Capacity and availability management activities typically involve:

- Developing a capacity and availability management approach and keeping it updated
- Providing and allocating resources
- Monitoring, analyzing, understanding, forecasting, adjusting, and reporting on current demand for:
 - Work activities

- Services
- Solutions and deliverables
- Resources
- Capacity and availability
- Service or service system performance
- Availability
- Determining corrective actions to ensure appropriate capacity and availability while balancing costs against resources needed and supply against demand

Explanatory PA Information

Practice Summary

Level 1

- PLAN 1.1 Develop a list of tasks.
- PLAN 1.2 Assign people to tasks.

Level 2

- PLAN 2.1 Develop and keep updated the approach for accomplishing the work.
- PLAN 2.2 Plan for the knowledge and skills needed to perform the work.
- PLAN 2.3 Based on recorded estimates, develop, and keep the budget and schedule updated.
- PLAN 2.4 Plan the involvement of identified stakeholders.
- PLAN 2.5 Plan transition to operations and support.
- PLAN 2.6 Ensure plans are feasible by reconciling estimates against capacity and availability of resources.
- PLAN 2.7 Develop the project plan, ensure consistency among its elements, and keep it updated.
- PLAN 2.8 Review plans and obtain commitments from affected stakeholders.

Level 3

- PLAN 3.1 Use the organization's set of standard processes and tailoring guidelines to develop, keep updated, and follow the project process.
- PLAN 3.2 Develop a plan and keep it updated using the project process, the organization's process assets, and the measurement repository.
- PLAN 3.3 Identify and negotiate critical dependencies.
- PLAN 3.4 Plan for the project environment and keep it updated based on the organization's standards.

Level 4

- PLAN 4.1 Use statistical and other quantitative techniques to develop and keep the project processes updated to enable achievement of the quality and process performance objectives.

Additional PA Explanatory Information

Planning activities can be performed at varying levels in an organization, e.g., enterprise, division, department, project, or workgroups; and therefore, can be performed by various groups in an organization. The organization defines its organizational structures and accompanying roles and responsibilities for the various functions throughout the organization, including the perspective for planning.

A “project plan” is the overall plan to manage the project, including a coherent picture of who does what and when they do it. Develop the project plan as a stand-alone document or distributed across multiple documents. Use it to track and communicate actual progress and to determine and take corrective action. As the project progresses, revise the project plan to address changes in requirements, commitments, inaccurate estimates, corrective actions, and methods of performing activities.

When planning resources, always consider capacity and availability. Capacity is the degree to which resources can support the work, while availability is the degree to which the resources are accessible when needed. Management of capacity and availability provides the sustained resources to meet requirements in a cost-effective manner.

Examples of capacity include:

- Number of people resources with specified skills
- Number and types of workstations
- Server capacity
- Network bandwidth
- Office space
- Tools

Examples of availability include:

- Business operating hours
- Service or system uptime
- Consumables, e.g., office supplies
- Staffing calendars
- Resources, e.g., personnel, tools
- Number of parking spaces

Related Practice Areas

Estimating (EST)

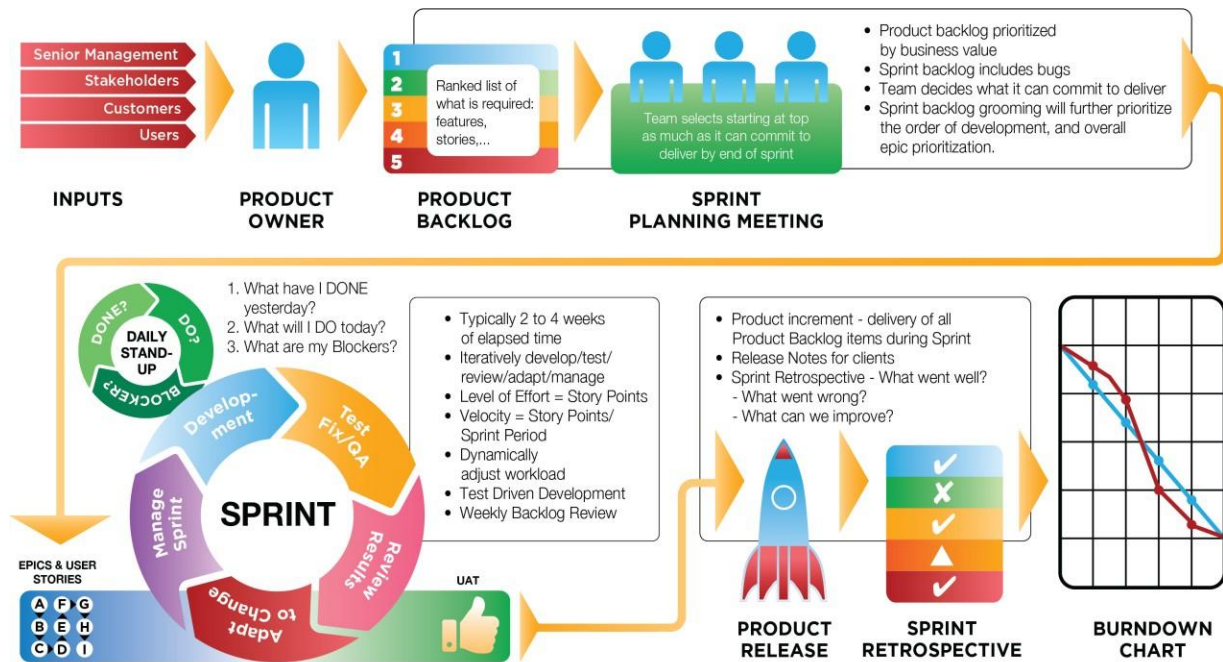
Monitor and Control (MC)

Risk and Opportunity Management (RSK)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

Figure PLAN-1: Agile Planning

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[Figure PLAN-1: Agile Planning](#) provides a summary view of a typical agile development workflow:

- **Product Owner:** The product owner gathers inputs from all affected stakeholders to organize and prioritize user stories into epics and Sprints to address the product backlog.
- **Product Backlog:** A prioritized collection of user stories, epics, and Sprints that represent the entire set of known requirements. Stories in the product backlog are often not yet estimated and are typically developed by the product owner, with assistance from business analysts and other team members.
- **Sprint Planning Meeting:** During this meeting, the collection of user stories or epics selected for the Sprint is estimated and further broken down into tasks by the agile team. The Sprint backlog is a forecast of what the team believes can be accomplished during the Sprint.
- **Team members “self-subscribe” to user stories and commit to completing them during the Sprint.** Responsibility for each story is usually recorded on a task board and can be redistributed as needed to manage the workload of the team. This information is used to provide input to schedule and budget planning.
 - **Schedule:** Each Sprint has a fixed duration, usually two to four weeks, and the collection of Sprints defines the total anticipated release schedule.
 - **Budget:** Agile teams typically adhere to a fixed team, fixed time-box model that, when examined in aggregate, helps to identify key drivers of the project’s budget.

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Define the data lifecycle that aligns to the business process, including traceability from the creation or acquisition of the data source to the target. Consider the following data lifecycle phases and scope the boundaries based on the approaches for managing data:

- Business Direction, e.g., data requirements, creation, acquisition
- Development, e.g., architecture, design
- Implementation, e.g., physical architecture
- Deployment, e.g., insertion into the operational environment
- Operations, e.g., data transformations, usage, performance, maintenance
- Retirement, e.g., decommissioning, archiving

Throughout the data lifecycle, as needed, obtain agreement from stakeholders regarding the data elements and authoritative data sources.

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Most engineering disciplines can benefit from project planning practices and include examples such as:

- Software development
- Hardware development
- Systems development
- Manufacturing or product lines, such as in:
 - Developing and maintaining core assets, e.g., components, tools, architectures, operating procedures, software
 - Supporting the use of core assets
 - Developing each individual system from core assets
 - Coordinating the overall effort of developing, using, and improving the core assets
- Construction and maintenance

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps teams build on agile planning principles, and in addition consider and address additional security requirements and constraints. DevSecOps is directly dependent on having consistent and repeatable agile development processes in place. One key consideration that should be addressed in planning is when developers will be using Free Open-Source Software (FOSS) as part of their software builds or incorporating third-party software in their product. If FOSS or third-party software is used, then planning should address:

- How developers are trained to protect the software they are writing from FOSS risks and supply chain attacks
- The mechanisms in place for developers to make them aware of existing vulnerabilities in a timely manner; this would include a list of the known vulnerabilities in the software they are planning to reuse or incorporate into their code, such as a Software Bill of Materials (SBOM)
- The processes and tools needed to detect and remove these vulnerabilities
- Identification of work arounds, mitigation strategies, and continuity response plans that can be implemented if an attack through one of these vectors should occur
- Any licensing or copyright considerations or restrictions on their use

People

Context Tag:	CMMI-PPL
Context:	Use processes to align workforce performance and business objectives.

Manage recruiting and retention by leveraging workforce competencies, including:

- Analyze recruitment and retention needs based on evaluation of the existing workforce against the workforce competencies
- Identify and analyze competency gaps, and compare with industry benchmarks
- Conduct recruiting or establish development plans based on analysis results
- Plan for personnel transitions to minimize disruption and maximize productivity

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Plan safety activities considering dependencies between safety components and any other activity that could be affected. Ensure that related impacts are identified and managed with planned safety activities. Integrate safety activities into the project plan and schedule, to the greatest extent possible, while taking into consideration requirements of regulatory agencies.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Determine the necessary security controls, based on the organization's security criteria. Identify and plan security activities, resources, and process assets, e.g., guidelines, templates, and tools; required to meet the necessary security level. This includes:

- Identify the necessary security posture
- Tailor the security activities and process assets for the work according to the security level
- Identify needed security resources and training to perform the work

Integrate the security activities and process assets within the overall work plan, including consideration of security resources and knowledge needed to perform the security activities.

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Responding to service requests may require detailed, frequently revised plans for allocating resources to tasks and managing the task queue, e.g., assigning repair jobs in a maintenance shop. Consider these low-level operating plans as an extension of the overall service plan. When planning for transitions in services or service system components, e.g., archival of a service, upgrade technology and account for all operational and support activities affected by the transition. Typically, organizations record this information in a transition plan. When planning for services, consider plans for service continuity.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Planning for supplier management is based on the acquisition strategy. The acquisition strategy is a guide for directing and controlling the work, and a framework for integrating activities essential to acquiring an operational solution. When needed, a supplier management approach should be developed and included as part of the planning activities and may include such items as:

- Availability of assets and technologies
- Supplier management objectives and constraints
- Consideration of supplier selection and management methods
- Potential supplier agreement types and terms

- Accommodation of end user considerations
- Consideration of supplier risk
- Support for the project supplier management activities throughout the project lifecycle

This involves the development and maintenance of plans for all supplier management processes, including plans required for effective acquirer-supplier interaction. Once the supplier agreement is signed and schedules, costs, and resources from the supplier are established, the acquirer takes the supplier estimations for the project into account, at an appropriate level of detail, in its project plan.

Planning for supplier management also includes establishing and keeping updated a plan for the orderly, smooth transition of the acquired product from a supplier to its use by the acquirer or the acquirer's customers. In addition, if an existing product is to be replaced as part of the acquired solution, the acquirer may be required to consider the disposal of the existing solution as part of the planning for acquiring the new solution. All transition activities are included in the project plan and provisions for accommodating such specialized requirements are also included.

Level 1

PLAN 1.1

Required Practice Information

Practice Statement

Develop a list of tasks.

Value

Ensures that the work needed to meet customer requirements is identified to increase customer satisfaction.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Develop a task list.	The level of detail of the task descriptions will vary based on the size and complexity of the tasks.
Review the task list with affected stakeholders.	
Revise the list as needed.	

Example Work Products

Example Work Products	Further Explanation
Task list	Includes: • Tasks to be performed and their timeline • Description of activities to assist in: ○ Assigning team members ○ Understanding what is expected

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

When providing a service, it is important to know and list what resources are available for performing the tasks needed to deliver the service. Identify needed and available resources to understand where the service may use resources. This can also help identify where shortages may occur. Unrecognized shortages may cause the organization to miss business objectives. Activities can include:

- Developing the task list
- Developing a list of needed resources by task
- Developing a list of the available resources

The task list may include a high-level description of the tasks that need to be performed to meet requirements or agreements. Include:

- Resource identifiers, e.g., a labor category for a human resource
- Descriptions of needed skills for the personnel resources, e.g., human resources, training, professional certifications
- Descriptions of needed equipment, tools, or facilities

PLAN 1.2

Required Practice Information

Practice Statement

Assign people to tasks.

Value

Ensures that tasks will be performed to meet requirements and satisfy the customer.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Ensure all tasks are assigned to specific members of the project and ensure that they know which tasks they are responsible for completing. Identify the needed skills, experience, abilities, and responsibilities needed to accomplish each task.

Example Activities

Example Activities	Further Explanation
Assign an individual who is responsible for each task.	Identify an individual with the skill and experience needed to accomplish the task.
Assign any additional people to the task.	Determine the project load and the abilities of the individuals as a part of assigning them to the task.
Review assignments with the assigned individuals.	Ensure that the assigned individuals understand what is needed to complete the task assignment.
Record assignments in task list.	Include feedback from assigned individuals on what is needed to accomplish the project.

Example Work Products

Example Work Products	Further Explanation
Task list with assignments	Includes: • Tasks • Description of each task • Name of the person or people assigned to work on that task

Level 2

PLAN 2.1

Required Practice Information

Practice Statement

Develop and keep updated the approach for accomplishing the work.

Value

Maximizes project success by keeping the affected stakeholders focused on accomplishing their specific objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

The approach is particularly valuable for making decisions when requirements and work changes. Identify the approach, and the priority of objectives for addressing requirements and accomplishing the work. Strategy and approach typically include:

- Business considerations
- Objectives and constraints
- Possible approaches to meeting those objectives and constraints
- Requirements
- Project lifecycle description
- Needed resources, e.g., skills, environment, tools, and new technologies
- Risks associated with the above and how they are mitigated
- Communication mechanisms, channels, and frequency

Example Activities

Example Activities	Further Explanation
Identify the objectives of the project.	Describe what the project is trying to accomplish in terms of expected outcomes and as applicable, Quality and Process Performance Objectives (QPPOs).
Identify the approach to be used to achieve objectives.	Describe the why and how: • Tasks are performed • The work is done • Methods or techniques are used • Resources are provided to support and accomplish the work

Example Activities	Further Explanation
	• Information, e.g., schedules, plans, budgets, and details on inputs and outputs, is communicated and made accessible
Identify requirements.	Record how requirements are addressed in the approach.
Record business considerations.	Business considerations may include: • Potential costs and benefits • Intellectual property • Competitive climate • Long-term needs and profit margins • Core competencies to be enhanced • Core competencies needed from other parties • Future trends
Define and record the project lifecycle.	The project lifecycle describes the major phases and activities needed to accomplish the work. The determination of a project's lifecycle phases provides for planned periods of evaluation and decision-making. These periods normally define logical points in the project plan, strategy, and approach to identify when decisions are needed to: • Stay on course • Change course including: ○ Reestimating cost and scope ○ Renegotiating commitments ○ Replanning resource usage ○ Adjusting budget and schedule • Stop work Understanding the project lifecycle is crucial in determining the scope of the planning effort and the timing of initial planning, as well as the timing and criteria (critical milestones) for replanning. Define the project lifecycle phases based on the requirements, the estimates for project resources, and the nature of the work.
Identify major resource needs and constraints.	Identify resource needs and the suppliers of these resources. For example: • Capacity and availability • Other groups in the organization • External suppliers
Identify affected stakeholders.	

Example Activities	Further Explanation
Record agreements with stakeholders.	Decide the nature of these agreements by considering each party's needs, objectives, expectations, constraints, and risks.
Identify risks or opportunities.	
Identify safety and security approaches.	Consider safety and security in all major planning activities.
Review the project approach with affected stakeholders and obtain agreement.	Review the approach from the following key business perspectives: • Are these the right objectives? • Is the approach feasible? • Will the project be subjected to excessive risk? • What safety and security issues are addressed and how well are they handled?
Revise the approach as necessary.	It may be necessary to refine the approach to reflect changes in the objectives, approach, availability of resources, market conditions, customer needs, technologies, etc.

Example Work Products

Example Work Products	Further Explanation
Recorded approach for accomplishing the objectives	
Recorded project lifecycle	

Related Practice Areas

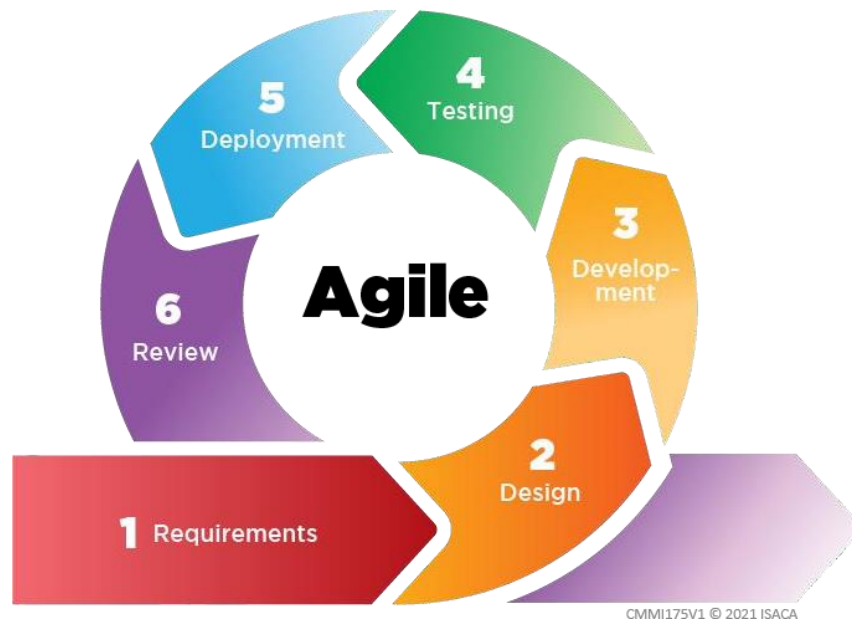
Risk and Opportunity Management (RSK)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

For agile projects, [Figure PLAN-2: Agile Development Lifecycle Phases](#) shows the most common phases of an agile development lifecycle. Each agile iteration follows these basic phases. In general, agile iterations or Sprints are two to four weeks.

Figure PLAN-2: Agile Development Lifecycle Phases**Development**

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Larger development projects can contain multiple phases, such as concept exploration, development, production, operations, and disposal. A development phase can include subphases such as requirements analysis, design, fabrication, integration, and verification. The determination of project phases typically includes selection and refinement of one or more development models to address interdependencies and appropriate sequencing of the activities in the phases.

Depending on the strategy for development, there can be intermediate phases for the development of prototypes, increments of capability, or spiral model cycles. In addition, include explicit phases for startup and close-out as needed.

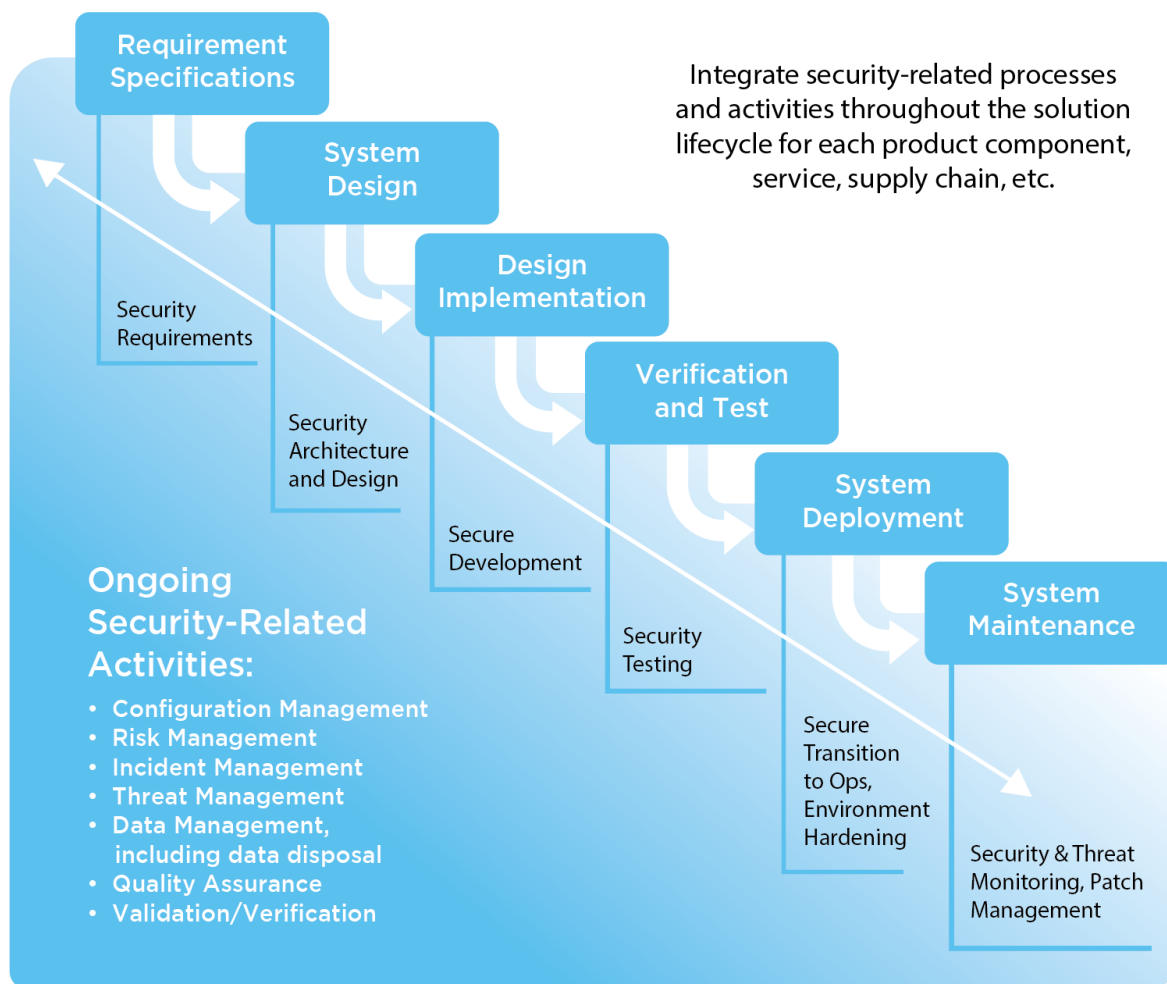
Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Most aspects of security are addressed multiple times within a solution lifecycle, but in general, the earlier in the lifecycle that security is addressed, the less effort and cost is ultimately required to achieve the same level of security. Software development in particular can introduce many security vulnerabilities. However, few software development methodologies or lifecycles explicitly address software security in detail, so secure software development practices must be added to each phase of the lifecycle to ensure that the software being developed is well-secured. For more information, refer to NIST Special Publication 800-218, Secure Software Development Framework.

[Figure PLAN-3: Development Lifecycle with Security Integration](#) provides an example of integrating security into a typical solution development lifecycle. While not all activities related to security are reflected, this graphic provides some of the most common considerations for integrating related processes. Additional security requirements and objectives should be considered during estimating, planning, and monitoring.

Figure PLAN-3: Development Lifecycle with Security Integration



Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

The service approach can play various roles, but initially, it serves as the basis for senior management to approve and commit resources. Revise the service approach through planning when identifying the solution, processes, resources, and risks.

A service approach can be developed by the organization, by prospective service personnel, e.g., in collaboration with potential customers and suppliers, or by some other combination of parties with a strategic business view of the service.

The service approach can include a high-level description of the service, the development approach, and the delivery approach and approached for automated service delivery, as appropriate.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquisition strategy defines the approach for formulating supplier request packages, supplier agreements, and plans. The strategy evolves over time and should continuously reflect the status and desired end of the work. The acquisition strategy establishes the acquisition phases, milestone decision points, and accomplishments for each acquisition phase. Develop the acquisition strategy from an understanding of the acquisition work and environment. The acquirer considers the:

- Potential value or benefit of the acquisition
- Risks or opportunities
- Constraints
- Experiences with different types of suppliers and agreements

The acquisition strategy includes:

- Acquisition objectives and constraints
- Asset and technology availability
- Consideration of acquisition methods
- Potential supplier agreement types
- Terms and conditions
- End user considerations
- Risk and opportunity considerations
- Core asset development and maintenance considerations
- Operational support processes

A well-developed acquisition strategy minimizes the time and cost required to satisfy approved capability needs and maximizes affordability throughout the lifecycle.

When an acquisition uses an evolutionary lifecycle, the strategy may describe the function, and how the acquirer and supplier will fund, develop, test, produce, and support the increasing functionality of the solution.

Business considerations for the acquisition strategy may include:

- Type of competition planned for all phases of the acquisition or an explanation of why competition is not practical or not in the best interests of the acquirer
- Developing or keeping updated access to competitive suppliers for critical solutions or solution components
- Availability and suitability of commercial items and the extent to which interfaces or connections for these items have broad market acceptance, standards, organization support, and stability
- Both international and domestic sources that can meet the required need consistent with organizational policies and regulations
- Critical technologies
- Data rights
- Product line considerations
- Socio-economic constraints
- Safety and health issues
- Security issues, including physical and information technology
- Other business-oriented solution quality characteristics that can be market differentiators or mission critical, e.g., solution responsiveness, platform openness, availability, sustainability

The acquisition strategy explains the planned acquisition incentive structure. This may include providing incentives for delivering the solution at or below the established cost targets, while satisfying requirements. Consider using incentives to manage risks.

If operations and maintenance support is going to be performed by an organization different from the supplier, define a sufficient overlap period to ensure a smooth transition. The acquirer's sustainment organization or supplier typically participates in the development of the solution support strategy.

PLAN 2.2

Required Practice Information

Practice Statement

Plan for the knowledge and skills needed to perform the work.

Value

Enables efficient and effective use of personnel resources.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Assess each individual's knowledge and skills against the critical skills required for work assignments to decide if training in these skills is needed. Examples of critical skills include the ability to:

- Execute specific processes
- Perform tasks within specific time limits
- Perform tasks to defined accuracy
- Use equipment safely and effectively
- Follow instructions
- Interpret information
- Organize actions, material, or people
- Communicate to perform assigned tasks. Examples of communication skills include:
 - Literacy in one or more of the languages used in the organization
 - Knowledge of local jargon or technical terms
 - Knowledge of communication protocols
 - Oral presentation skills
 - Negotiating skills
 - Writing skills
 - Ability to use communication media

Examples of items to be considered when evaluating knowledge and skills may include:

- Individual's personal assessment
- Individual's previous experience
- Performance feedback sessions and reviews
- Tests
- Training records

Example Activities

Example Activities	Further Explanation
Identify the knowledge and skills needed to perform the work.	
Determine the gaps between the knowledge and skills needed versus those held by the currently assigned people.	Consider current and future needs when identifying gaps. Succession planning is often used as a strategy for retaining capabilities.

Example Activities	Further Explanation
Select methods for providing needed knowledge and skills.	Training may be either in-house (both organizational and project) or external. Methods may include: • Staffing and new hires • Classroom instruction • Online or computer-aided training • Apprenticeship or mentorship programs • Job rotation or cross-training • Conferences, seminars, workshops, and tutorials • College and university courses • Videos • Directed self-study courses
Incorporate selected methods into the project plan.	

Example Work Products

Example Work Products	Further Explanation
Inventory of skill needs	Results may include the following needs: • Skills • Experience • Training

Related Practice Areas

Organizational Training (OT)

Workforce Empowerment (WE)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

For long-duration and continuous-operation services, the knowledge and skills needed change as:

- Personnel rotate in and out of the projects (or from one service type to another)
- Technology used in the service system or for an individual service change
- Processes and technology used in the development or customer environments change

For example, a personnel change triggers the need to reevaluate the knowledge and skills required for new team members. The types of knowledge and skills needed may change during different phases of the service lifecycle, when adding new services, or changing service levels. Plan for needed knowledge and skills to better address these sources of change.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer plans for knowledge and skills required by the acquisition team to perform their tasks.

For example, if the acquirer is purchasing a software-intensive solution, ensure that the assigned acquisition personnel have expertise in systems and software engineering, or the acquirer will train personnel in these areas. The acquirer requires that personnel have orientation and training in acquirer processes and domain knowledge. Personnel involved in receiving, storing, using, and supporting the acquired solution also may need appropriate training.

The acquirer also plans for knowledge and skills needed from the supplier. For example, the acquirer can provide role descriptions and skill profiles to the supplier as part of the supplier request package.

PLAN 2.3

Required Practice Information

Practice Statement

Based on recorded estimates, develop, and keep the budget and schedule updated.

Value

Enables timely management and corrective actions needed to achieve objectives, through early detection of significant deviations from the budget and schedule.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Record and keep updated the budget, effort, and duration needed to accomplish the work.

Example Activities

Example Activities	Further Explanation
Identify major milestones.	Milestones can be event-based or time-based.
Identify schedule assumptions.	Assumptions are frequently made on items with little, if any, available estimation data. Identifying assumptions about activity duration provides insight into the confidence level, i.e., uncertainties, in the overall schedule.

Example Activities	Further Explanation
Identify constraints.	Identify factors that limit planning flexibility as early as possible. Examining solution and task characteristics identifies these issues. Constraints may include: • Customer requirements • Resources • Defining dependencies among activities (predecessor or successor relationships) • Supplier availability
Identify task dependencies.	A critical part of ordering tasks for a project is identifying task dependencies. Tools and inputs that can help determine the optimal ordering of task activities may include: • Critical Path Method (CPM) • Program Evaluation and Review Technique (PERT) • Resource limited scheduling • Customer priorities • User or end user value • Work packages
Identify resources.	Project resources may include: • Staffing needs and costs ◦ Base personnel decisions on projects, tasks, roles and responsibilities, and the knowledge and skills required for each position • Equipment • Environment • Materials • Facilities • Consumables • Access to intellectual property • Transportation, e.g., people and materials
Identify and analyze risks.	Review and analyze identified constraints and record them as risks if they potentially impact the budget or schedule.
Based on estimates, develop the budget and schedule, and keep them updated.	Often includes: • Estimating size, complexity, effort, duration, and cost • Defining the committed or expected availability of resources • Determining the time phasing of activities • Determining a breakout of subordinate schedules • Defining schedule activities of appropriate duration • Identifying releases or increments for the delivery of solutions

Example Activities	Further Explanation
	• Using appropriate historical data to verify the schedule • Defining incremental funding requirements • Updating project assumptions and rationale • Updating risks • Defining a management reserve based on the risks to the schedule and budget
Establish corrective action criteria.	Establish criteria to determine what constitutes a significant deviation from the project plan. A basis for gauging issues and problems is necessary to decide when to take corrective action.

Example Work Products

Example Work Products	Further Explanation
Budget	The budget should be based on tasks and resources.
Schedule	The schedule should be based on tasks, resource availability, and dependencies. The schedule and Work Breakdown Structure (WBS) should be aligned to the project lifecycle. May include a: • WBS or list of tasks and activities • Task dictionary
Resource plan	May include: • Personnel requirements based on work size and scope • Critical facilities and equipment list
Budget and schedule risks	

Related Practice Areas

Estimating (EST)

Risk and Opportunity Management (RSK)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Individual service requests, e.g., to repair a piece of equipment in a remote facility, transport a package to a destination, can have individual milestones, task dependencies, resource allocations, and scheduling constraints. Consider these milestones together and coordinate with larger budgeting and scheduling activities.

Develop communication methods during service system development. Review, tailor, and possibly supplement communication methods regularly to meet ongoing service delivery needs.

Budget development for service-related activities may include:

- Service components consumed during the lifecycle of a service request or for multiple service requests that span multiple customers
- Type of resources required, e.g., 24-hour available resources for VIP services
- Expected number of service requests considered in a service agreement
- New resources that need to be added, purchased, or modified to comply with current or future demand

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

For the duration of the work, define, track, and keep updated:

- Budget and schedule, including both acquirer and supplier activities
- Critical dependencies of supporting organizations, including any suppliers that support the acquirer

PLAN 2.4

Required Practice Information

Practice Statement

Plan the involvement of identified stakeholders.

Value

Ensures that stakeholder needs are addressed when they arise, reducing the amount and cost of rework.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Stakeholder involvement incorporates ongoing sharing of information and allows the project to focus on:

- Developing the right solutions
- Delivering adequate services
- Acquiring compatible solutions that provide the required functionality

Identify personnel or workgroups who must be included in the project for it to be accomplished in accordance with requirements. Affected stakeholders may include:

- People and functions that should be represented
 - Those affected by the activity
 - Those who have the needed expertise to perform the activity

Affected stakeholders are those who will participate in the project's activities. The list of affected stakeholders may change as work progresses. It is important to ensure that affected stakeholders have early input to the planning of the project.

Some stakeholders, such as higher levels of management and customers, may have no formal role in the project, but are invested in the outcome. Consult these stakeholders for advice and consent when the scope may change.

Key elements of effective information sharing include:

- Ensuring the right set of people receive the information
- Sharing information in a timely manner
- Clarifying, as needed, to ensure that the information is understood
- Following up, as needed, to ensure that the information is being acted on

Examples of methods to share information required for performing work include:

- Meetings, e.g., in-person meetings, video conferences, and teleconferences
- Online systems, e.g., ticketing systems, workflow systems, requisitioning systems, and databases
- Online collaboration tools
- Emails

Example Activities

Example Activities	Further Explanation
Develop a list of stakeholders.	
Identify the involvement of each stakeholder.	Describe how and why each stakeholder is involved and what is required of them.
Record when the involvement is required.	Address the timing and sequencing for stakeholder input and involvement.

Example Activities	Further Explanation
Identify information to be shared on an ongoing basis.	Consider the following aspects of the communication: • Information needs • Sources of the information, e.g., status, client, and updates • Responsibilities for communications, e.g., individuals and workgroups
Identify the communication parameters.	Consider the following aspects which influence and dictate the types of communication: • Mechanisms, including technology considerations • Scope • Importance of message • Channels • Frequency • Security • Privacy • Confidentiality
Share information and collect any communication feedback.	Use feedback to improve future communications.

Example Work Products

Example Work Products	Further Explanation
Stakeholder involvement plan	May include: • List of affected stakeholders • Rationale for involvement • Relationships, roles, and responsibilities • Schedule of interaction • Communication strategy
Responsible, Accountable, Supporting, Consulted, Informed (RASCI) table	Defines the RASCI for activities and work products.
Results of information sharing	For example, meeting minutes, data in online systems, and emails.

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Plan for the involvement of stakeholders with service system transition activities, e.g., inclusion of a new product, update of technology for service system components. Affected stakeholders include customers, end users, provider personnel, senior management, external suppliers, and anyone else who should be aware of expected changes. Consider the magnitude of the change in planning for the appropriate level of detail and involvement during the transition.

Example mechanisms for involving stakeholders of service system transitions include:

- Automatic notifications from the service system
- Posted information within the service system
- Progress reviews, discussions, or approaches
- Communications through newsletters
- Updates of user guides and user procedures
- Updates of user training

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Stakeholders can include acquirers, supplier team members, end users, and other involved parties. When acquiring complex integrated solutions involving multiple suppliers, the acquirer needs to ensure coordination between all stakeholders. This planning typically includes steps for developing and keeping updated supplier agreements with these stakeholders, e.g., interagency and intercompany agreements, memoranda of understanding, memoranda of agreement.

PLAN 2.5

Required Practice Information

Practice Statement

Plan transition to operations and support.

Value

Minimizes surprises and rework during adoption and deployment.

Additional Required Information

Planning for transition should be considered part of the initial planning for the project. Even in the case of small projects, transition and delivery of the solution must be addressed.

Explanatory Practice Information

Additional Explanatory Information

Plans for the transition to operations and support include the:

- Approach for introducing and maintaining readiness
- Sustainment

- Operational capability of the solutions
- Assignment of responsibility for the transition, delivery, and support
- Activities needed to manage the transition, e.g., project kickoff
- Personnel, including staff augmentation
- Support for the solution in its intended environment, including:
 - Definition of transition readiness criteria
 - Reviews with affected stakeholders
 - Safety considerations
 - Security considerations
- Handling of potential risks, including safety risks, security threats, and vulnerabilities
- Changes to the solution over time
- Eventual removal of the solution from operational use

Example Activities

Example Activities	Further Explanation
Determine transition scope and objectives.	
Determine transition requirements and criteria.	Includes timing, dependencies, and other high-level elements of moving the solution into operations.
Determine approach to transition.	
Develop schedule for transition.	
Determine transition responsibilities and resources including post-transition support.	If the project team is not responsible for transition, described who is responsible. Includes: • Transition infrastructure • Future enhancements
Determine operations and support training needs.	Identifies any additional skills, experience, or knowledge needed for successful and smooth operation.

Example Work Products

Example Work Products	Further Explanation
Plans for transition to operations and support	May include: • Scope and objectives • Assignment of responsibility • Transition processes and procedures • Activities needed to manage the transition and support the solution in its intended environment

Example Work Products	Further Explanation
	• Risks • Evaluation methods and acceptance criteria to ensure the transition of the solution to operations and support • Readiness criteria for the operations organization and environment • Transition of intellectual property or other assets to the designated repository • Resolution steps if problems are encountered • Readiness criteria for the solutions • Readiness criteria for the solution support organization • Identification of the maintenance organization

Context Specific

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Identify and plan the security activities, resources, and process assets, e.g., specifications, templates, and tools; required to ensure that a consistent security level is obtained and preserved during operations and transition to operations.

The plan must address:

- Stakeholder responsibilities
- Security procedures
- Security characteristics of the intended operational environment, e.g., physical access, networks, and perimeter protection
- Specific information necessary to perform security activities

Examples of security assets to consider are:

- Security patch management procedures, which includes communication, delivery, and installation
- Security configuration and hardening guidelines, which includes scope and content of the configuration; and hardening for the different solution components, which includes third-party components, if appropriate
- Security delivery and operations instructions, which includes information about maintaining the security level during operations, e.g., for managing roles and privileges, secure handling of passwords, security incident handling, and usage of digital signatures or fingerprints
- Security commissioning instructions, which includes information for addressing security during commissioning, e.g., for changing default passwords

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

If support will be provided by an organization different from the supplier, a sufficient overlap period should be included in the plan.

Typically, the acquirer develops initial transition and support plans and then reviews and approves more detailed transition and support plans.

In an acquisition, the transition and support plan should also include:

- Expectations for supplier execution of the transition
- Warranty expectations for the acquired solution
- Transition of intellectual property or other acquirer assets to the acquirer's designated repository
- Resolution steps if any problems are encountered

PLAN 2.6

Required Practice Information

Practice Statement

Ensure plans are feasible by reconciling estimates against capacity and availability of resources.

Value

Increases likelihood that the objectives are achieved by ensuring that needed resources are available and committed to throughout the project.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Overcommitted resources can be reconciled by:

- Resource leveling the schedule
- Modifying or deferring requirements
- Negotiating more resources
- Finding ways to increase productivity
- Modifying or negotiating the scope of work involved, such as phased delivery
- Outsourcing
- Adjusting the personnel skill mix
- Revising all plans that affect the project or its schedules

- Identifying tools, techniques, and methods that could reduce time or cost
- Implementing incremental delivery
- Renegotiating stakeholder commitments

Manage individual project assignments to balance committed work among individuals and projects:

- Evaluate individual workloads periodically to ensure they are balanced. Adjust individual commitments as needed to improve balance and avoid over commitment.
- When individuals' work is nearing completion, seek opportunities to apply their effort to other business activities.
- Ensure the manager of an individual committed to work on several projects:
 - Ensures the combined commitments do not result in over commitment
 - Coordinates expectations for timing of work and results
 - Resolves conflicts among project commitments

Example Activities

Example Activities	Further Explanation
Identify and plan for resource capacity and availability.	
Perform resource leveling to adjust scheduling of tasks and resources.	Resource leveling balances the demand for resources with their availability.
Ensure commitments are supported by adequate personnel or other required resources.	
Negotiate commitments with affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Revised plan and commitments	

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

While selecting a supplier and negotiating the supplier agreement, the acquirer reconciles overall work and resource levels based on supplier proposals. After completing the supplier agreement, the acquirer incorporates the supplier's plans at an appropriate level of detail to support plan alignment. For example, an acquirer can incorporate major supplier milestones, deliverables, and reviews.

The resource plan should plan for personnel with the appropriate training and experience to evaluate supplier proposals and participate in negotiations with suppliers. The resource plan identifies the work resources expected from the supplier, including critical facilities or equipment needed. Revise the resource plan based on the supplier agreement or changes in environment conditions.

PLAN 2.7

Required Practice Information

Practice Statement

Develop the project plan, ensure consistency among its elements, and keep it updated.

Value

Ensures efficient and effective communication and achievement of objectives through a consistent project plan.

Additional Required Information

A plan addresses the planning and management of work activities collectively. A plan is more than just a list of scheduled activities and tasks. It includes an approach for how the work is to be performed and managed. Developing and using a project plan includes these iterative activities:

- Identifying required tasks, including the activities and their interdependencies to be used for tracking and managing the project
- Using estimates to:
 - Develop a budget and schedule
 - Determine the resources required
- Determining the tasks and their interrelationships
- Assigning roles and responsibilities for performing the tasks
- Identifying realistic performance data and objectives:
 - Determining if objectives can be achieved as the project progresses
 - Evaluating alternative objectives when the objectives cannot be met
- Identifying and analyzing risks or opportunities
- Determining information and data management needs and how they will be addressed
- Identifying and interacting with affected stakeholders to adequately address technical and support activities in project plans

- Negotiating and obtaining commitment to the project plan

Explanatory Practice Information

Additional Explanatory Information

A project plan can include multiple plans. The plan generated for the project defines all aspects of the effort, tying together the following in a logical manner:

- Tasks
- Budgets and schedules
- Milestones
- Information management and governance
- Risks
- Resources and skills
- Stakeholder roles and involvement
- Infrastructure
- Improvement and sustainment activities

A plan provides the basis for:

- Monitoring and managing project performance
- Taking timely corrective actions to ensure the objectives are met
- Replanning when needed

Example Activities

Example Activities	Further Explanation
Record the project plan.	
Review the project plan with affected stakeholders.	Ensure that the project plan describes a realistic approach for meeting the needs, expectations, and constraints of affected stakeholders and helps ensure that these affected stakeholders will fulfill their roles.
Revise the project plan as necessary.	Planning is an iterative activity.

Example Work Products

Example Work Products	Further Explanation
Overall project plans	A project plan may be composed of multiple plans (which may be separate or combined into one or more documents) such as: • Staffing Plan • Stakeholder Involvement Plan • Data Management Plan • Data Migration Plan • Communication Plan

Example Work Products	Further Explanation
	• Performance Improvement and Sustainment Plan • Measurement and Analysis Plan • Monitoring and Control Plan • Solicitation and Supplier Management Plan • Risk and Opportunity Management Plan • Transition Plan • Quality Management Plan • Configuration Management Plan • Security Plan • Safety Plan • Workgroup Management Plan

PLAN 2.8

Required Practice Information

Practice Statement

Review plans and obtain commitments from affected stakeholders.

Value

Reduces rework and increases the likelihood of achieving objectives through a consistent understanding and commitment to the plan.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

All plans that affect the project should be reviewed to obtain a common understanding of the scope, objectives, roles, responsibilities, and relationships. Ensure that individuals or groups making a commitment are confident that the project can be performed within cost, schedule, and performance constraints.

Example Activities

Example Activities	Further Explanation
Ensure individuals are involved in reviewing the work they are responsible for and the inputs that initiate the work.	Review the decisions, agreements, and related information needed to understand requirements and plans necessary for accomplishing the work.
Record commitments.	• A commitment is a pact that is freely assumed, visible, and expected to be kept by all involved

Example Activities	Further Explanation
	• Ensure individuals and workgroups make commitments for work they will be accountable for performing • Commitments can be internal and external • Record commitments to ensure a consistent mutual understanding and for project tracking and maintenance. Provisional commitments may include a description of risks associated with the relationship.
Review and approve project commitments.	• Work with the appropriate levels of management as defined by the process • The plan for stakeholder interaction should identify all parties from whom commitment should be obtained

Example Work Products

Example Work Products	Further Explanation
Results of plan reviews	Reviews will help identify issues that lead to misunderstandings or may prevent objectives from being met. Review results may include: • List of issues discovered during review • List of changes that will be made • Reason for plan changes
Recorded commitments	Includes decisions and agreement to requirements, project plans, and their related elements.

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Development activities in preparation for production can be included in the hardware development plan or defined in a separate production plan.

Examples of plans that have been used by large organizations include:

- Integrated Master Plan – an event driven plan that records significant accomplishments with pass/fail criteria for both business and technical elements of the project and that ties each accomplishment to a key project event.
- Integrated Master Schedule – an integrated and networked multi-layered schedule of project tasks required to complete the work recorded in a related Integrated Master Plan.

- Systems Engineering Management Plan – a plan that details the integrated technical effort across the project.
- Systems Engineering Master Schedule – an event-based schedule that contains a compilation of key technical accomplishments, each with measurable criteria, requiring successful completion to pass identified events.
- Systems Engineering Detailed Schedule – a detailed, time dependent, task-oriented schedule that associates dates and milestones with the Systems Engineering Master Schedule.

Level 3

PLAN 3.1

Required Practice Information

Practice Statement

Use the organization's set of standard processes and tailoring guidelines to develop, keep updated, and follow the project process.

Value

Establishes the project process, ensuring the efficient and effective achievement of the objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Select and tailor or modify processes from the organization's set of standard processes to build a process to fit the specific needs of the project. The project process is tailored from the organizational set of standard processes in accordance with organizational tailoring guidelines, and:

- Includes a process description that is kept up-to-date
- Contributes to organizational assets

A project's process is based on:

- Stakeholder requirements
- Goals and objectives
- Work product and task characteristics
- Impacts or requirements of the lifecycle
- Support activities
- Commitments
- Organizational process needs and objectives
- The organization's set of standard processes and tailoring guidelines
- Domain
- The operational environment
- The business environment
- Stakeholder availability
- Experience of the people
- Project constraints

When the organization's set of standard processes change, there may be changes in the project's processes.

Example Activities

Example Activities	Further Explanation
Select standard processes from the organization's set of standard processes that best fit the needs of the project.	Collectively, these comprise the project process
Modify the organization's set of standard processes and other organizational process assets according to tailoring guidelines to produce the project's process.	
Use other artifacts from the organization's process asset library as appropriate.	Other artifacts can include: • Estimating models • Lessons learned documents • Templates • Example documents
Record the project process.	The project process covers the activities for the work and interfaces or connections to affected stakeholders.
Review the project process.	Record and use review results, inputs, and issues of the project processes to identify potential impacts.
Revise the project process as necessary.	As the project progresses, revise the description of the project process to better meet project requirements and the organization's process needs and objectives.

Example Work Products

Example Work Products	Further Explanation
The project process	Describe how the organizational processes are implemented for this specific project, including any unique, more detailed, or additional processes or procedures that are needed and allowed by the organizational tailoring guidelines.

Related Practice Areas

Process Asset Development (PAD)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

The description of the defined process should be based on services that the project delivers, including both tailored standard services and unique services, and the service delivery environment.

Organizations that define standard services may have individual systems that enable the delivery of those standard services. Any processes that are components of service system(s) within an organization are good candidates to consider when defining the standard processes for delivering services.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The supplier agreement should identify the key activities to deliver a solution that meets requirements. The acquirer may require the supplier to align selected supplier processes with the acquirer's defined process to achieve the project goals.

The acquirer should align its defined process with the acquisition strategy. For example, whether the acquisition strategy is to introduce new technology or to consolidate existing solutions affects the acquirer's defined process.

Supplier deliverables may include:

- Common defined processes
- Requirements tools shared by both acquirer and supplier
- Test tools and facilities shared by both acquirer and supplier

PLAN 3.2

Required Practice Information

Practice Statement

Develop a plan and keep it updated using the project process, the organization's process assets, and the measurement repository.

Value

Increases the likelihood that the objectives will be met, by using proven organizational assets for planning the project.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Addresses organizational learning and enhances performance improvement by providing proven assets to projects, which gives them the best chance for success.

Developing a project plan and keeping it updated should address additional planning activities such as:

- Incorporating the project process
- Coordinating with affected stakeholders
- Using organizational process assets
- Incorporating plans for peer reviews
- Establishing objective entry and exit criteria for tasks

Examples of data contained in the organization's measurement repository which can be used for planning include:

- Size
- Effort
- Cost
- Schedule
- Staffing
- Response time
- Supplier performance
- Defects or issues

Example Activities

Example Activities	Further Explanation
Use the tasks and work products of the project process as a basis for estimating and planning project activities.	
Use the organization's measurement repository to estimate the work.	Estimates and supporting information typically include: • Validated historical data from this project or similar projects • Similarities and differences between the current project and the work represented by the historical data • Reasoning, assumptions, and rationale used to select historical data
Integrate or develop plans.	Other plans that affect the project plan can include: • Staffing plans • Training plans • Stakeholder involvement plans

Example Activities	Further Explanation
	• Performance improvement and sustainment plans • Measurement and analysis plans • Monitoring and control plans • Solicitation plans • Agreement management plans • Risk mitigation plans • Transition plans • Quality assurance plans • Configuration management plans • Verification and validation plans • Peer review plans
Incorporate the definitions of measures and measurement activities.	Measures may include the: • Organization's common set of measures • Additional project and product specific measures
Establish objective entry and exit criteria for tasks and activities.	Entry and exit criteria make it clear when people are needed and when they are to start and complete their tasks.
Identify how conflicts will be resolved that arise among affected stakeholders.	Identify resolution approaches and mechanisms for addressing and resolving conflicts, including any agreed to escalation procedures.

Example Work Products

Example Work Products	Further Explanation
Revised project estimates	Estimates that are based upon organizational experience with other similar projects help to accomplish work more accurately and effectively.
Project plans	
Integrated plans	Describe how plans interact and are integrated with each other, including the dependencies, sequences, inputs and outputs of the plans and tasks and how they relate to each other.

Related Practice Areas

Estimating (EST)

Managing Performance and Measurement (MPM)

Risk and Opportunity Management (RSK)

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Examples of development-specific parameters that are considered for similarities and differences include:

- Design and development approaches

Examples of development-specific product and project interface or connection risks include:

- Incomplete interface or connection descriptions
- Unavailability of commercial off-the-shelf (COTS) components

An additional development-specific example of scheduling factors includes:

- Integration and test issues

Development activities in preparation for production can be included:

- in a software or hardware development plan
- in a system engineering management plan
- in a separate production plan

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

The work plan should include service delivery and, when needed, service system development and service system transition.

Examples of the types of service data contained in the organization's measurement repository include:

- Service capacity
- Number of service requests received, closed, cancelled, or in progress
- Number of service requests that missed their service level agreement
- Standard services from the catalog that are in high demand
- Average service request completion time
- Average cost consumed by service request

Plans that affect the work plan include:

- Capacity and availability management strategy
- Service continuity plan
- Incident management approach
- Risk and opportunity management approach

When developing a plan for the project, it is important to develop, keep updated, and follow a strategy for capacity and availability management for critical services. This will help ensure the right resources will be available to meet the requirements when needed and that the organization can meet commitments to customers within agreed limits.

A capacity and availability management strategy is based on requirements, failure and incident trend analysis, current resource use, and service system performance. Strategy should consider the minimum, maximum, and average use of resources over the short, medium, and long term.

It may be appropriate for some projects to identify, plan for, and manage the availability of resources to respond to sudden, unexpected increases in demand. Sometimes, the management of the obsolescence of certain resources and offerings factor into the strategy for capacity and availability management.

Service representations can help to determine resources and aspects to measure, monitor, analyze, and manage. However, solution design documents may not be available or may not accurately and completely reflect all aspects of the live environment that affect capacity and availability. It is important to monitor and analyze actual capacity and availability data. Requirements strategies, monitoring, and information from day-to-day service delivery, product development, or acquisition can assist with these determinations.

The capacity and availability management strategy can reflect limits and constraints, e.g., limited customer funding, the customer's acceptance of certain risks related to capacity and availability.

The provider must formulate a strategy that best meets requirements, even if they cannot influence or control demand and resource adjustments. This strategy can be more sophisticated in situations where the provider can influence or control demand and resource adjustments.

Example activities for developing a capacity and availability management strategy include:

- Recording actual resource use, performance, and availability data
- Estimating future resource and capacity and availability requirements
- Developing a capacity and availability strategy that meets requirements, meets the demand for resources and services, products, or acquisitions, and addresses how the organization provides, uses, and allocates resources
- If appropriate, including an availability testing schedule, a maintenance strategy, and an outage schedule
- Recording costs and benefits of the strategy and any assumptions
- As necessary, revising the strategy on an event-driven basis

PLAN 3.3

Required Practice Information

Practice Statement

Identify and negotiate critical dependencies.

Value

Reduces risk and increases the likelihood the project will be completed on time, within budget, and meet quality objectives by paying close attention to critical dependencies.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

This section left blank for future content.

Example Activities

Example Activities	Further Explanation
Identify each critical dependency.	Record dependencies, relationships, and how they affect the plan. Consider personnel or staff augmentation dependencies.
Integrate schedules.	An integrated master schedule (IMS) can be used to identify critical dependencies between work groups and functions.
Review and negotiate dependencies with affected stakeholders.	Communicate new or existing dependencies and changes with affected stakeholders to ensure that they can perform their work.
Record commitments to address each critical dependency.	

Example Work Products

Example Work Products	Further Explanation
Critical dependencies	May include: • Description of the critical dependencies • Commitments made during negotiations • Risks associated with the dependency

PLAN 3.4

Required Practice Information

Practice Statement

Plan for the project environment and keep it updated based on the organization's standards.

Value

Ensures that the resources needed to complete the work are readily available to maximize productivity.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Describe the project environment to address any unique or critical aspects that may affect the project being performed.

An appropriate project environment is comprised of an infrastructure of facilities, tools, and equipment that people need to perform their jobs effectively in support of business and project objectives. Keep the project environment updated at a level required by the organizational project environment standards. Develop or acquire the project environment or its components.

Example Activities

Example Activities	Further Explanation
Analyze project and plan for resources, facilities, and environment	The critical aspects of the project environment are requirements driven. Explore project environment functionality and quality characteristics with the same rigor as any other project planning activity. Examples of the resources to be considered include: • Considerations for the project environment ○ Cultural issues ○ Visibility ○ Voice communication ○ Comfortable furniture ○ Airborne particulates ○ Light for performing work ○ Isolation and noise protection ○ Space to perform group work activities ○ Meeting space ○ Support areas ○ Laboratories • Specialized characteristics of workspaces ○ Safety ○ Security ○ Air conditioning ○ Tools ○ Training ○ Remote locations ○ Telecommuting • Storage

Example Activities	Further Explanation
	• Project equipment ○ Computers or workstations ○ Communications technologies, e.g., telephones, fax machines, email, and networks ○ Office equipment ○ Printing and reproduction equipment • Supplies • Application software • Documentation
Assign a responsible individual to plan to acquire resources, facilities, and the environment needed to perform assigned work.	Examples of actions include: • Preparing budget requests • Developing cost-benefit justifications • Consulting with Subject Matter Experts (SMEs) • Submitting purchase orders • Negotiating with those responsible for managing building or computing facilities, distributing equipment or supplies, or other project environment-related resources
Develop a contingency plan if the resources, facilities, and environment cannot be obtained.	
Plan for support personnel needs.	Examples may include: • Business or administrative • Computer support personnel • Technical writing and documentation • Laboratory technicians
Ensure individuals and groups provide input and participate in decisions concerning resources, facilities, and environment.	Inputs may be taken on a periodic basis, e.g., annually through employee engagement surveys; or based on a significant event, e.g., moving to a new facility, changing working hours, or introducing a personnel canteen. Consider the following areas for collecting inputs: • Arrangement of project facilities • Alterations or improvements to the project environment • Resources needed to perform their project activities • Remote work operational parameters Examples of mechanisms for gathering opinions from individuals include: • Opinion surveys or organizational culture and engagement questionnaires • External surveys or benchmark reports • Interviews with members of the workforce

Example Activities	Further Explanation
	• Discussions with management, including meetings that allow individuals to skip levels of management or to meet with management representatives, such as an ombudsman • Group meetings on concerns • Focus groups or advisory boards • Postmortem project reviews • Suggestion boxes or other private means of accepting comments • Email or other electronic means To maximize benefits from inputs: • Acknowledge the receipt of the opinions • Analyze the inputs collected to identify action plans for implementation • Share the analysis along with actions identified for implementation with the members of the workforce • Implement the actions identified • Keep the workforce informed about the status of the identified actions
Track actions identified and communicate status.	

Example Work Products

Example Work Products	Further Explanation
Project resources, facilities, and environment plan	May be included as part of the overall project plan or may be separate, e.g., for large or complex projects or environments. Include approach for collecting and addressing workforce and user inputs.
Equipment and tools for the project	
Health and safety considerations, including any regulatory or legal requirements	
Installation, operation, and maintenance manuals for the project environment	
User surveys and results	
Facilities, resources, and maintenance records for the project	
Support services needed for the project environment	

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

The project environment might encompass environments for product development, integration, verification, and validation, or they might be separate environments.

Development-specific examples of equipment and tools include:

- Design tools
- Configuration management tools
- Evaluation tools
- Integration tools
- Automated test tools

Components in the development environment include software, databases, hardware, tools, test equipment, and appropriate documentation. Qualification of software includes appropriate certifications. Hardware and test equipment qualification includes calibration and adjustment records and traceability to calibration standards.

Examples of actions to improve the development environment include:

- Adding new tools
- Acquiring additional networks, equipment, training, and support

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Consider security needs when planning for the project environment. Key activities include:

- Analyzing technology needs, e.g., review current platforms, servers, networks, and tools; and conduct a security needs assessment
- Identifying criteria for selecting security tools and methodologies
- Evaluating and selecting security tools
- Applying security tools and methodologies
- Defining physical security processes, procedures, and protocols

Consider an environment implementation that incorporates defense in depth to maximize overall mitigation of security risk. Consider the following types of security tools when planning for the secure environment:

- Badging and physical access control
- Common Weakness Enumeration (CWE) scans, to help identify software vulnerabilities

- Secure coding standards and guidelines
- Attack modeling methods/tools
- Source code security analyzers, or Static Application Security Testing (SAST) tools

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Verification and validation of the service system can include both initial and ongoing evaluation of the environment in which the service provider delivers the service.

Components in the services work environment include those necessary to support service delivery, software, databases, hardware, tools, test equipment, and appropriate documentation. Qualification of a service delivery environment includes audits of the environment and its components for compliance with safety requirements and regulations. Software qualification includes appropriate certifications. Hardware and test equipment qualification includes calibration, records and traceability to calibration standards.

Examples of actions to take to improve the services work environment include:

- Adding new equipment and tools
- Acquiring additional networks, equipment, training, and support

Examples of equipment and tools include:

- Resource management tools for service delivery
- Incident and request management tools

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Ensure that the supplier's and acquirer's projects are compatible to enable the efficient and effective transfer of work products and solutions.

Define the project environment in the plan, and include any environments required throughout the project lifecycle.

Example environment types include:

- Acquirer or supplier facilities
- Independent Verification and Validation (IV&V)
- Configuration Management
- Testing
- Infrastructure hosting
- Information repositories
- Field sites
- Classified facilities

Level 4

PLAN 4.1

Required Practice Information

Practice Statement

Use statistical and other quantitative techniques to develop and keep the project processes updated to enable achievement of the quality and process performance objectives.

Value

Increases the likelihood that the processes of the project will enable achievement of consistent performance and quality.

Additional Required Information

Managing the progress towards achieving Quality and Process Performance Objectives (QPPOs) should be an integral part of how the project is planned and managed.

Explanatory Practice Information

Additional Explanatory Information

Ensure that projects:

- Identify the processes required to accomplish business activities.
- Identify conditions that may affect performance of the processes.
- Identify relevant process performance baselines and models for the selected processes.
Projects may have historical process performance baselines and models that are more accurate or relevant than organizational baselines.
- Evaluate the planned performance of the selected processes to determine if they are capable of achieving the measurable performance objectives.
- When planned performance is not being achieved, negotiate adjustments in measurable performance objectives and the associated processes. This may involve identifying and developing new processes or subprocesses.

Example Activities

Example Activities	Further Explanation
Develop the criteria to use in evaluating process alternatives for the project.	Criteria can be based on: • QPPOs • Availability of process performance data and the relevance of the data to evaluating an alternative • Previously recorded process performance baselines and models that can be used in evaluating an alternative • Lifecycle models

Example Activities	Further Explanation
	• Stakeholder requirements • Laws and regulations
Identify or develop alternative processes for doing the work and meeting objectives.	May include: • Analyzing organizational process performance baselines and models to identify candidate processes • Identifying processes from the organization's set of standard processes as well as tailored processes in the process asset library • Identifying processes from external sources, e.g., other organizations, professional conferences, academic research • Adjusting the level or depth of intensity with which the processes are applied and can involve the: ○ Number, type, and date of peer reviews to be held ○ Amount of effort or calendar time devoted to particular tasks ○ Number and selection of people involved ○ Skill level requirements for performing specific tasks ○ Selective application of specialized construction or verification techniques ○ Reuse decisions and associated risk mitigation strategies ○ Sample size for peer reviews
Analyze and evaluate alternative processes against recorded evaluation criteria.	Analyze the relative strengths and weaknesses of alternatives. This analysis can be supported by aligning the organization's process performance models with process performance data. Perform additional modeling activities if existing process performance models cannot address significant relationships among the alternative processes under consideration and there is high risk of not achieving objectives. Use historical data and process performance baselines and models to assist in evaluating alternatives against the criteria. These evaluations can include a sensitivity analysis, particularly in high-risk situations.
Select the alternative process that best meets the criteria.	If needed, repeat these activities several times until confident that the best available alternative has been identified.
Evaluate the risk of not achieving the project's QPPOs.	If the risk cannot be avoided or mitigated, it may be necessary to revise the project's QPPOs.

Example Work Products

Example Work Products	Further Explanation
Criteria used to evaluate alternatives for the project	
Alternative processes	It may be possible to develop more than one candidate defined process for the project. In this case, select the one that best meets the evaluations criteria.
Selected defined project process	
Risk assessment of not achieving the project's QPPOs	Risk handling plan and action to be identified. Decision analysis may be used to identify alternate risk mitigation practices.

Related Practice Areas

Decision Analysis and Resolution (DAR)

Managing Performance and Measurement (MPM)

Risk and Opportunity Management (RSK)